

Hackathon Project Phases Template

Project Title:

CodeGenie: AI-Powered Code Generation using CodeLlama

Team Name: TECH AI

Team Members:

- Chilakala Manikanta Sai Anurudh
 - Konagalla Abhirama Rao
 - Katakam Revanth Kumar
 - Edulamudi Manoj
 - Bandaru Anil Kumar
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Phase-1: Brainstorming & Ideation

Objective:

Identify the problem statement, define the purpose and impact of the project.

Key Points:

1. **Problem Statement:** Developers often spend a significant amount of time writing repetitive code, debugging, and understanding complex code structures. CodeGenie aims to simplify and accelerate the coding process using AI-powered code generation
2. **Proposed Solution:** CodeGenie leverages CodeLlama to generate optimized code snippets, provide detailed explanations, and assist in debugging. It streamlines development by reducing coding errors and saving time.
3. **Target Users:**
 - Software developers

- Students and educators
 - Tech startups
 - Open-source contributors
4. **Expected Outcome:** A fully functional AI-powered coding assistant that helps developers write, understand, and debug code faster, enhancing productivity and efficiency.

Phase-2: Requirement Analysis

Objective:

Define technical and functional requirements for the CodeGenie: AI-Powered Code Generation using CodeLlama

Key Points:

1. Technical Requirements:

- Programming Languages: Python (for backend), JavaScript (for frontend)
- Frameworks: Flask/Django (backend), React.js (frontend)
- AI Model: CodeLlama (for code generation)
- APIs: OpenAI API (if needed), GitHub API (for code suggestions)
- Database: PostgreSQL/MySQL

2. Functional Requirements:

- Code generation based on text prompts
- Syntax highlighting and error checking
- Code explanation feature
- Debugging assistant
- Multi-language support

3. Constraints & Challenges: (Any limitations or risks)

- Ensuring accurate and optimized code generation
- Handling edge cases in different programming languages
- Latency and response time optimization
- Maintaining security and preventing AI-generated vulnerabilities

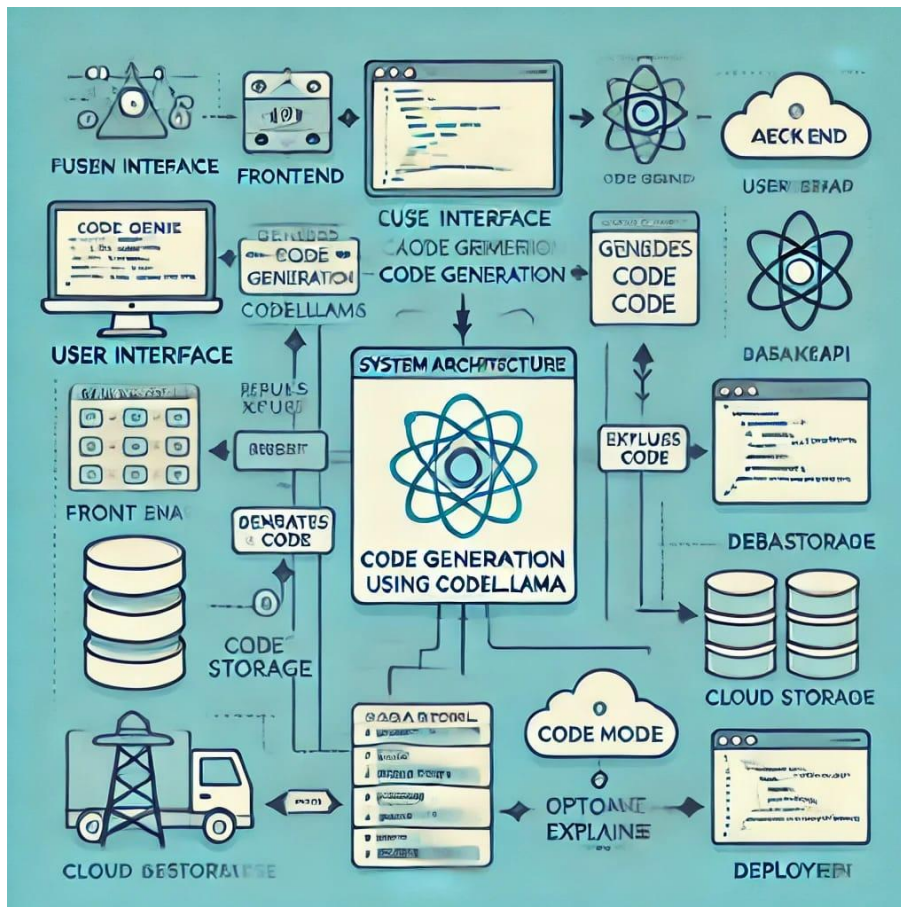
Phase-3: Project Design

Objective:

- Create the architecture and user flow.

Key Points:

1. System Architecture Diagram:



2. User Flow:

1. User enters a prompt describing a function or task.
2. CodeGenie processes the request and queries CodeLlama.
3. AI generates relevant code along with explanations.
4. The user can review, edit, or request modifications.
5. Debugging suggestions are provided if needed

3. UI/UX Considerations:

- A minimalistic interface with a code editor-like experience
 - Dark/light mode support
 - Interactive response for debugging assistance
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Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Key Points:

1. Sprint Planning:

sprint	Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
Sprint 1	Environment Setup & API Integration	 High	6 hours (Day 1)	End of Day 1	Revanth Kumar	CodeLaama API key, Pytho, Fast API set up	API connection established & working
Sprint 2	Implementing core AI-based code generation	 Medium	2 hours (Day 1)	End of Day 1	Abhirama Rao	CodeLlama model set up, API response format finalized	Basic UI with input fields for code generation
Sprint 3	Adding debugging and explanation features	 High	3 hours (Day 2)	Mid-Day 2	Manoj & Manikanta	API response processing, UI elements ready, LLM responses for matting	Code Debbuging assistance with explanation
Sprint 4	Testing and refining UI/UX	 High	1.5 hours (Day 2)	Mid-Day 2	Anil & Revanth	API logs, UI Responsivene s testing	Improved API stability and smooth UI interactions

2. Task Allocation:

Backend: kongalla Abhirama rao
Frontend: Edulamudi Manoj
AI Model Integration: Katakam Revanth Kumar
Testing & Debugging: Bandaru Anil Kumar

3. Timeline & Milestones:

Day 1: Initial setup and model integration
Day 1: Core features implementation
Day 2: UI development and debugging features
Day 2: Final testing and deployment

Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

1. **Technology Stack Used:**
 - Backend: Python (Flask/Django)
 - Frontend: React.js
 - AI Model: CodeLlama
 - Database: PostgreSQL
 2. **Development Process:**
 - Setting up API calls for CodeLlama
 - Creating endpoints for code generation and debugging
 - Implementing a frontend UI for user interaction
 - Testing and optimizing AI responses
 3. **Challenges & Fixes:**
 - Setting up API calls for CodeLlama
 - Creating endpoints for code generation and debugging
 - Implementing a frontend UI for user interaction
 - Testing and optimizing AI responses
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Phase-6: Functional & Performance Testing

Objective:

- Ensure the project works as expected.

Key Points:

1. **Test Cases Executed:**
 - Code generation for different programming languages
 - Error handling in syntax and logic
 - Performance testing for response time
2. **Bug Fixes & Improvements:**
 - Fixed incorrect function outputs
 - Optimized model response time
3. **Final Validation:**
 - Ensured the AI generates relevant, executable code
 - Verified debugging suggestions are accurate

4. **Deployment :**

- Hosting on AWS/GCP
- Creating a demo environment

Final Submission

1. Project Report Based on the templates
2. Demo Video (3-5 Minutes)
3. GitHub/Code Repository Link: <https://github.com/Manikanta904/Techai.git>
4. Presentation